MOTIONIX

Updated Connections for 315MHz RF Receiver with GND, 5V, D0, D1, D2, D3, VT Terminals

Yes, theseterminalsare correctand standardfor a 315MHz4-channelRFreceivermodule(likePT2272-based ones). Here's what each does (quick explanation for clarity):

GND: Ground connection (common negative).

5V: Power supply (VCC)—this module requires 5V (not 3.3V like some others), so we'll connect it to our 5V source.

D0, D1, D2, D3: Data outputs—each goes HIGH (5V) when the corresponding button on the remote is pressed (D0 for Button 1, D1 for Button B/2, etc.).

VT: Valid Transmission output—goes HIGH briefly when any valid signal is received (useful for LED indicator or signal confirmation, but optional in our setup).

Important Updates Based on These Pins:

Voltage Adjustment: The receiver runs on 5V (from our DC-DC buck converter), not ESP32's 3.3V. Data pins output 5V logic, which could damage ESP32 (max 3.6V tolerant). So, I've added simple voltage dividers (using 10kΩ and 4.7kΩ resistors per data pin) to step down 5V to ~3.3V—safe and easy.

VT Usage: We'll connect it optionally to an LED (via 330Ω resistor) for visual signal confirmation, but you can skip if not needed.

Remote Mapping: As per your instructions: Button 1 (D0) = Arm/Disarm; Button B/2 (D1) = Self-Start; Button C/3 (D2) = Horn/Alarm; Button 4 (D3) = Headlight.

Safety Note: All updates keep things safe—no shorts. Use the buck converter's 5V for receiver power. Test receiver alone with 5V first (press remote buttons—measure ~5V on D pins).

The rest of the system (power, ESP32, MPU6050, relays, etc.) remains unchanged. Here's the ***full, complete connection list*** in simple, step-by-step format—wire in this order.

1. Power & Charging Circuit Connections

Vivo Li-ion Battery (+) terminal → First B+ terminal on TP4056

Vivo Li-ion Battery (-) terminal → First B- terminal on TP4056 + Common ground rail (black wire)

Micro-USB cable 5V (+) → IN+ terminal on TP4056

Micro-USB cable GND (-) → IN- terminal on TP4056

Second B+ terminal on TP4056 (output) → AMS1117 Regulator VIN terminal

Second B- terminal on TP4056 (output) → Common ground rail

AMS1117 Regulator VOUT terminal (3.3V) → Schottky Diode cathode terminal (striped end)

Schottky Diode anode terminal → ESP32 VIN pin

12V Adapter (+) terminal → DC-DC Buck Converter VIN terminal

12V Adapter (-) terminal → Common ground rail

DC-DC Buck Converter 5V OUT+ terminal → RF Receiver 5V terminal + 4-Channel Relay Module VCC terminal

- DC-DC Buck Converter GND terminal → Common ground rail

2. ESP32 Power & Ground Connections

ESP32 VIN pin ← Schottky Diode anode terminal (~2.85V-3.3V safe)

ESP32 GND pin → Common ground rail

ESP32 EN pin → 10kΩ resistor one end → ESP32 3.3V pin (other end of resistor)

ESP32 3.3V pin → MPU6050 VCC terminal (sensors get 3.3V from here)

3. RF Receiver (315MHz 4-Channel) Connections (Updated for Your Terminals)

RF Receiver 5V terminal → DC-DC Buck Converter 5V OUT+ terminal (powers at 5V)

RF Receiver GND terminal → ESP32 GND pin / Common ground rail

RF Receiver D0 terminal (Button 1) → Voltage Divider: Connect D0 to 10kΩ resistor → Junction → ESP32 GPIO2 pin; Junction to 4.7kΩ resistor → GND (steps down 5V to ~3.3V safe)

- RF Receiver D1 terminal (Button B/2) → Voltage Divider: D1 to 10kΩ → Junction → ESP32 GPIO4 pin; Junction to 4.7kΩ → GND

RF Receiver D2 terminal (Button C/3) → Voltage Divider: D2 to 10kΩ → Junction → ESP32 GPIO5 pin; Junction to 4.7kΩ → GND

RF Receiver D3 terminal (Button 4) → Voltage Divider: D3 to 10kΩ → Junction → ESP32 GPIO18 pin; Junction to 4.7kΩ → GND

RF Receiver VT terminal → Optional: Connect to 330Ω resistor → LED Anode; LED Cathode → GND (LED lights on any signal for testing; skip if not needed)

RF Receiver ANT terminal (if present, though not listed) → 22cm wire antenna for range

(Voltage Divider Note: For each D pin, the 10kΩ/4.7kΩ combo divides 5V to ~3.3V—safe for ESP32. Solder on breadboard. Total cost: ₹20 for resistors.)

4. Motion Sensor (MPU6050) Connections

MPU6050 VCC terminal → ESP32 3.3V pin

MPU6050 GND terminal → ESP32 GND pin

MPU6050 SDA terminal → ESP32 GPIO21 pin

MPU6050 SCL terminal → ESP32 GPIO22 pin

5. 4-Channel Relay Module Connections

Relay Module VCC terminal → DC-DC Buck Converter 5V OUT+ terminal

Relay Module GND terminal → ESP32 GND pin

Relay Module IN1 terminal → ESP32 GPIO13 pin (controls Relay 1 - Ignition)

Relay Module IN2 terminal → ESP32 GPIO14 pin (controls Relay 2 - Self-Start)

Relay Module IN3 terminal → ESP32 GPIO15 pin (controls Relay 3 - Horn/Alarm)

Relay Module IN4 terminal → ESP32 GPIO16 pin (controls Relay 4 - Headlight)

6. Ignition Demo Circuit (Relay 1 + Key Switch + Motor 1)

Motorcycle Key Switch Terminal 1 → 12V Adapter (+) terminal

Motorcycle Key Switch Terminal 2 → Relay Module COM1 terminal

Relay Module NO1 terminal → 12V DC Motor 1 (+) terminal (ignition demo)

12V DC Motor 1 (-) terminal → 12V Adapter (-) terminal

7. Self-Start Demo Circuit (Relay 2 + Motor 2)

Relay Module COM2 terminal → 12V Adapter (+) terminal

Relay Module NO2 terminal → 12V DC Motor 2 (+) terminal (self-start demo)

12V DC Motor 2 (-) terminal → 12V Adapter (-) terminal

8. Horn/Alarm Circuit (Relay 3 + Horn)

Relay Module COM3 terminal → 12V Adapter (+) terminal

Relay Module NO3 terminal → Horn (+) terminal (horn demo/alarm)

Horn (-) terminal → 12V Adapter (-) terminal

9. Headlight Demo Circuit (Relay 4 + LED)

Relay Module COM4 terminal → 12V Adapter (+) terminal (or ESP32 3.3V for low-power)

Relay Module NO4 terminal → LED Anode terminal (via 330Ω resistor)

LED Cathode terminal → 12V Adapter (-) terminal (or ESP32 GND)

10. Battery Voltage Monitoring Connections

100kΩ Resistor one end → Second B+ terminal on TP4056

100kΩ Resistor other end → Junction point (connect to 27kΩ resistor and ESP32 GPIO34 pin)

27kΩ Resistor one end → Junction point

27kΩ Resistor other end → ESP32 GND pin

11. Safety & Filtering Components

470µF Capacitor (+) → DC-DC Buck 5V OUT+ terminal → (-) to ground (filter relay power)

470µF Capacitor (+) → Second B+ terminal on TP4056 → (-) to ground (filter battery output)

100nF Capacitors (x6): Place one across VCC/GND of ESP32, RF Receiver, MPU6050, Relay, and motors (noise reduction)

10kΩ Pull-up Resistors (x4): One each on ESP32 GPIO2, GPIO4, GPIO5, GPIO18 (to ESP32 3.3V to prevent floating signals)

This updated list incorporates your receiver's exact terminals while keeping everything safe (e.g., voltage dividers prevent 5V damage to ESP32). Test the receiver with 5V power first—press remote buttons and

measure ~5V on D pins (then add dividers). If you need code adjustments or more details, let me know! 🚀

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